



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014

BIOLOGY
Higher 1
Paper 2 Core Paper

8875/02
Thursday, 28 August
2 hours

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows an electron micrograph of part of a liver cell.

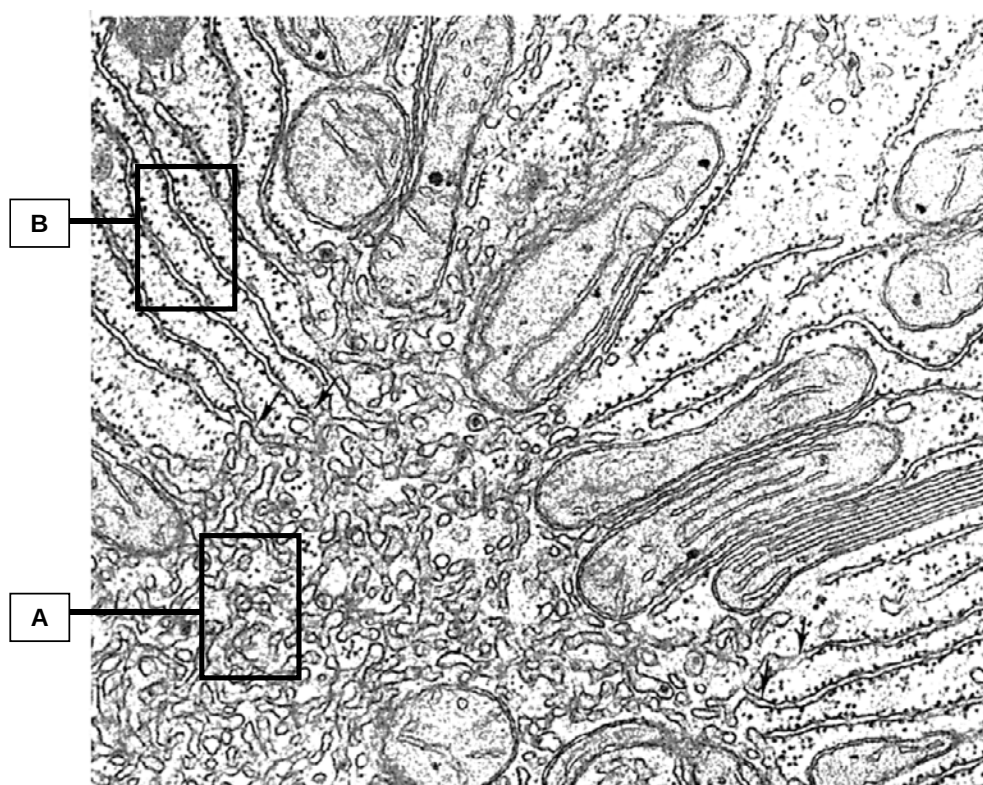


Fig. 1.1

- (a) Identify structures A and B. [2]

A: smooth endoplasmic reticulum

B: rough endoplasmic reticulum

(b) State the function of structure A. [1]

Concerned primarily with the synthesis of lipids (including membrane phospholipids), cholesterol and steroid hormones.

(c) Relate the structural features of B to its function. [2]

1a) Structure: Ribosomes are present on its surface.

1b) Function: This is the site where proteins are made and transported through the cisternae of the ER.

2a) Structure: B consists of a network of membranous tubules and sacs / called cisternae.

2b) Function: Transportation of the protein product takes place: through the cisternae to its destination, then packaged in membranous vesicles and exported / dispatched to other parts of the cell

Cholesterol is taken up by liver cells, which transforms it into 30 to 100 nm low-density lipoprotein (LDL) particles. Fig. 1.2 shows the structure of an LDL particle. It consists of a core of triglycerides and a surface coat of phospholipids, cholesterol and a monomeric protein, ApoB-100.

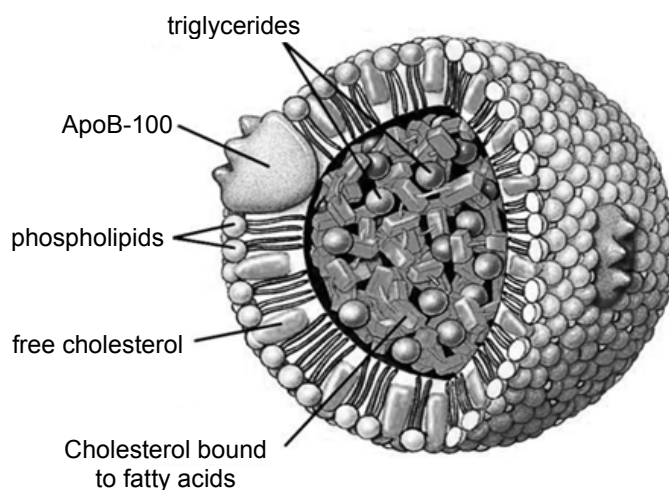


Fig. 1.2

(d) With reference to Fig. 1.2, outline the roles of lipids in the LDL coat. [3]

Cholesterol and phospholipids are important components of the LDL coat:

1. Free cholesterol – maintains fluidity of the LDL coat.
2. It disturbs the close packing of phospholipids thus keeping them more fluid.
3. Phospholipids – major constituent of the LDL coat, they form the monolayer in LDL coat.

4. The role of the phospholipid monolayer is to form a boundary between the hydrophobic triglyceride core from the aqueous exterior of the LDL.
5. The hydrophobic tails face inwards and make contact with the hydrophobic triglyceride core
6. The hydrophilic phosphate heads face outwards and make contact with the exterior aqueous environment.

(e) Fig. 1.3 shows the three dimensional structure of ApoB-100.

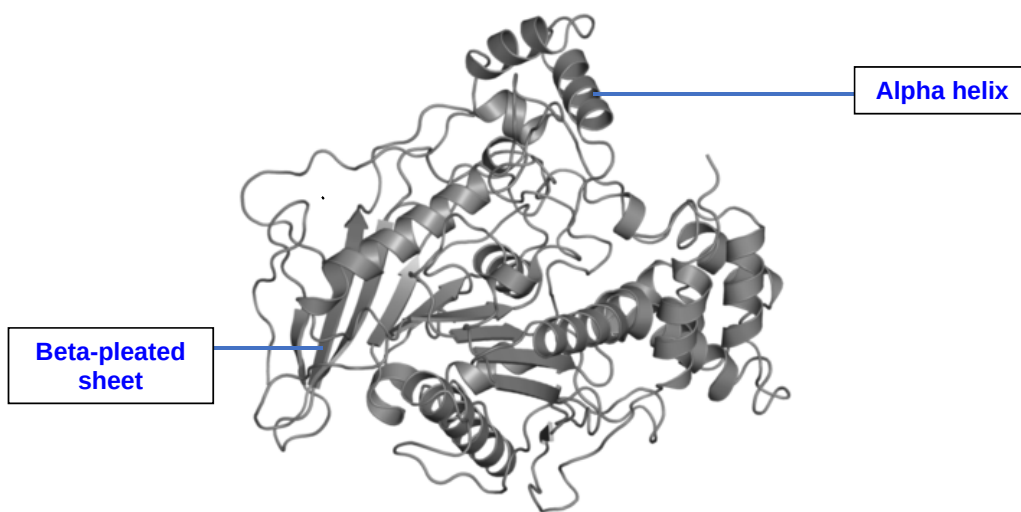


Fig. 1.3

(i) Label the type(s) of secondary protein structure observed in Fig. 1.3. [1]

(ii) Explain how the structure of ApoB-100 protein is maintained. [3]

1. The primary sequence of the ApoB-100 protein is held by peptide bonds between amino acid residues;
2. The secondary structure consisting of the α -helices and β -pleated sheets
3. are maintained by hydrogen bonds between peptide linkages of the polypeptide backbone – the R groups of amino acids are not involved.
4. The tertiary structure of the ApoB-100 protein is maintained by the interactions among the R groups of amino acids at different regions in the polypeptide chain –
5. namely the disulphide bond, the hydrogen bond, the ionic bond and hydrophobic interactions.
6. ApoB-100 folds into a compact, globular 3-dimensional structure.

[Total: 12]

- 2 In DNA replication, primers are added, then progressively removed and replaced by DNA nucleotides for both the leading strand and the lagging strand.

- (a) State **one** similarity and **one** difference between the synthesis of the leading strand and the lagging strand during DNA replication. [2]

Similarity: (any one)

1. Both synthesized from 5' to 3' direction / reading of the template strand is from 3' to 5' direction.
2. Both require DNA polymerase and primer for synthesis of the new strand.
3. Both strands are formed by the addition of free deoxyribonucleotides via complementary base pairing to the template strand.

Difference: (any one)

1. The synthesis of the leading strand only requires one primer, while the lagging strand requires many primers.
2. The synthesis of the leading strand is continuous, while that of the lagging strand is discontinuous / occurs in Okazaki fragments.
3. The synthesis of the leading strand does not require DNA ligase, while that of the lagging strand requires ligase to join the Okazaki fragments / fragments
4. The leading strand grows towards the replication fork as it is synthesized while the lagging strand grows away from the replication fork.

- (b) Give **one** reason why the primer removed from the 5' end of the daughter strand is **not** replaced by DNA nucleotides. [1]

1. DNA polymerase can only add nucleotides to the 3' end of an existing nucleotide strand.
2. This is because the shape of active site of DNA polymerase is complementary to the shape of the 3'-OH group of an existing nucleotide strand.

The replication of DNA requires enzymes and substrates. One of the substrates is ATP.

However it has been found that 9- β -D-Arabinofuranosyladenine (ara-ATP) is a chemical that affects the activity of an important enzyme in the process.

In an investigation, the effect of different concentrations of ATP on the rate of DNA synthesis was determined:

- with no ara-ATP
- with a low concentration of ara-ATP
- with a high concentration of ara-ATP

The results of the investigation are shown in Fig. 2.2.

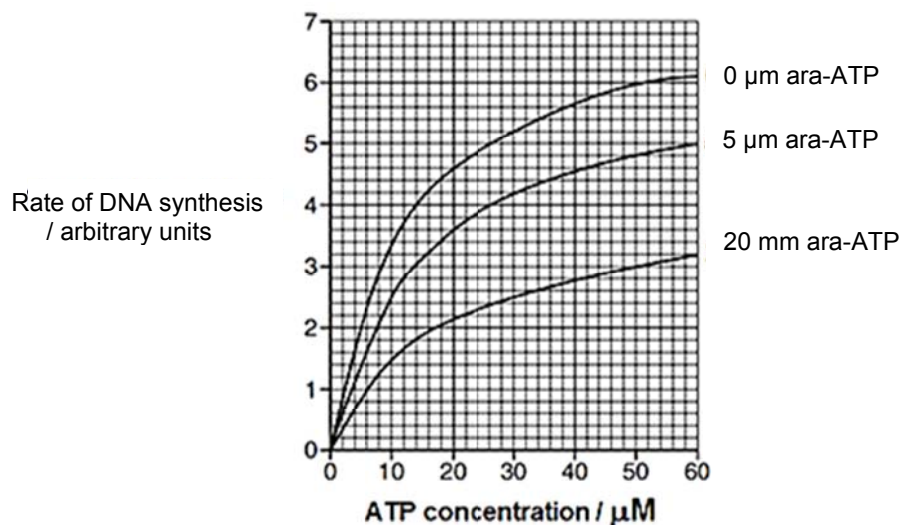


Fig. 2.2

(c) Explain the effects of ara-ATP on the rate of DNA synthesis. [4]

1. G.S : Increase in concentration of ara-ATP concentration resulted in a decrease in rate of DNA synthesis for all concentrations of ATP.
2. Q. F. : At low ATP concentration of 10 μM , 0 μM of ara-ATP resulted in a rate of 3.4 a.u. of DNA synthesis while 5 μM of ara-ATP resulted in a rate of 2.5 a.u. of DNA synthesis
 [Student to choose any two ara-ATP conc to compare]
 [Student may choose 20 μM of ATP for low ATP conc.]
3. Q. F. : At high ATP concentration of 60 μM , 0 μM of ara-ATP resulted in a rate of 6.1a.u. of DNA synthesis while 5 μM of ara-ATP resulted in a rate of 5 a.u. of DNA synthesis
 [Student to choose any two ara-ATP conc to compare]
 [Student to choose from 40 μM ATP onwards for high ATP conc.]
4. Rate of DNA synthesis is an indication of the rate of enzyme activity.
5. ara-ATP acting as a non-competitive inhibitor of the enzyme.
6. It binds to the enzyme at a site other than the active site

7. Cause a change in the conformation of the enzyme active site therefore substrate unable to bind with active site / fewer enzyme-substrate complexes are formed
8. Therefore rate of reaction of DNA synthesis will not reach max even at high substrate conc.

(d) State **two** ways in which polymerase chain reaction (PCR) is similar to the replication of DNA. [2]

Any two:

1. In both processes, the DNA molecule need to unwind / the DNA molecule need to be separated into 2 strands before the process can occur.
2. In both processes, free deoxyribonucleotides are matched to the template via complementary base pairing
3. In both processes, DNA polymerase enzymes are required in both processes.
4. In both processes, the DNA strands are synthesized in the 5' to 3' direction.
5. In both processes, both involve semi-conservative replication whereby the daughter DNA molecule consists of 1 parental strand and 1 daughter strand.
6. In both processes, both parent strands act as template
7. In both processes, primers are needed to initiate the process.

[Total: 9]

- 3 Fig. 3.1 shows the pedigree chart of a family with congenital alactasia, also known as congenital lactase deficiency. It is a disorder in which infants are unable to break down lactose in milk and can result in severe diarrhoea, dehydration and weight loss.

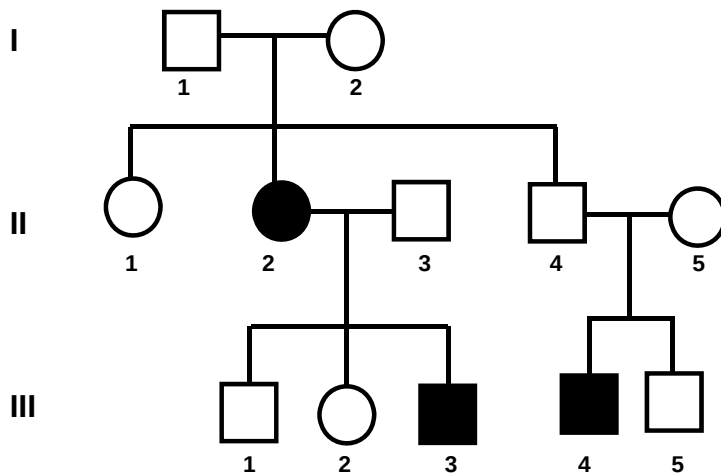


Fig. 3.1

- (a) With reference to Fig. 3.1, explain the mode of inheritance of the allele for congenital alactasia.

1. This disease is determined by a autosomal recessive allele
2. Trait skips generations
3. Unaffected parents(I-1 and I-2) have affected children (II-2)/
affected child (III-4) has unaffected parents (II-4 and II-5)/OWTTE

[2]

- (b) Using suitable symbols, state the possible genotype(s) of individuals II-2 and III-5.

II-2 : aa

III-5 : Aa, AA

[1]

The *LCT* gene codes for lactase. This enzyme helps to digest lactose, a sugar found in milk and other dairy products. At least nine *LCT* gene mutations cause congenital lactase deficiency (congenital alactasia). The *LCT* gene mutations either change the polypeptide sequence of lactase or result in an enzyme that is abnormally short.

- (c) Describe how a gene mutation can result in an enzyme that is abnormally short.

1. Substitution of a single base/Inversion of two bases could also lead to a stop codon being present.
2. This could lead to premature termination of translation, giving rise to a truncated protein.

OR

1. If the mutation was a single base insertion or deletion and leads to a frame-shift mutation.
2. The reading frame was completely changed from the point of addition/deletion which may consist of a premature stop codon resulting in the production of a truncated protein.

[2]

The domestication of plants and animals roughly 10,000 years ago profoundly changed human diets. In Europe, the domestication of cattle brought about an increase in nutrition from cow's milk.

Several thousand years later, a genetic mutation spread through Europe that gave people the ability to produce lactase. This adaptation opened up a rich new source of nutrition that could have sustained communities when harvests of crops failed.

At present, more than 90 % of adults in Britain and Scandinavia have the ability to break down lactose found in milk.

(d) Explain how the process of natural selection could have resulted in a large proportion of adults in Europe with the ability to break down lactose.

1. Europeans would have varying abilities to break down lactose/Some Europeans have the ability to produce lactase
2. due to mutations that has spontaneously occurred over time.
3. the failure of harvest/availability of food could have been the selection pressure
4. Individuals who are able to break down lactose/produce lactase have a selective advantage
5. They will live to reproductive age to produce viable offspring
6. Able pass on the favourable alleles to the next generation
7. An increase in frequency of favourable alleles
8. Natural selection would operate, leading to an increase in frequency of individuals with the ability to break down lactose and a decrease in frequency of the other individuals.

[4]

[Total: 9]

- 4 Photosynthetic and metabolic acclimation to low temperatures were studied in Arabidopsis. A group of plants were grown at 23 °C and then shifted to 5 °C (i.e. 23/5 °C). The leaves were collected and frozen at the temperature of liquid nitrogen. Enzyme activity in the leaves was then compared to those grown in 23 °C and in 5 °C.

The activity of the C3-cycle enzymes (Rubisco, PGK and GAPDH) is presented in Fig. 4.1.

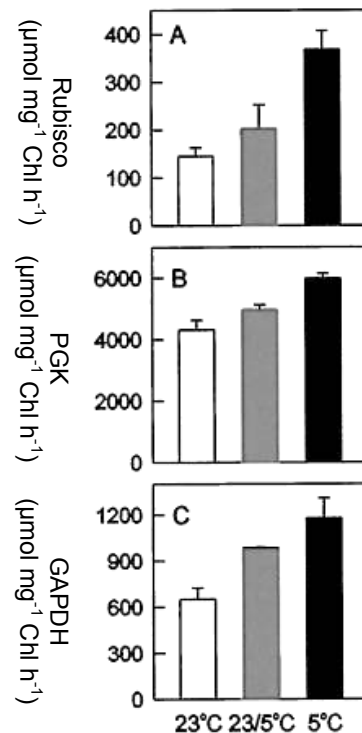


Fig. 4.1

(a) Compare the effect of temperature on the activity of the C3-cycle enzymes. [2]

1a. At higher temperature of 23 °C, enzyme activity of the C3-cycle enzymes was lower than at 5 °C and at 23/5°C

OR the reverse: enzyme activity was higher at 5 °C than at 23 °C and at 23/5°C

1b. For example, activity of Rubisco at 23 °C was only 150 μmol mg⁻¹ Chl h⁻¹, as compared to 200 μmol mg⁻¹ Chl h⁻¹ at 23/5°C and 360 μmol mg⁻¹ Chl h⁻¹ at 5 °C/ any other valid QF

2a. Temperature change had the greatest effect on Rubisco than on PGK and GAPDH

2b. (QF any relevant data, e.g.) change in temperature from 5 °C to 23 °C showed a decrease from 360 to 150 μmol mg⁻¹ Chl h⁻¹ in Rubisco activity, as compared to a decrease from 6000 to 4200 μmol mg⁻¹ Chl h⁻¹ in PGK and from 1200 to 620 μmol mg⁻¹ Chl h⁻¹ in GAPDH.

- (b) State the role of Rubisco in the C₃-cycle. [1]
 1. Enzyme involved in CO₂ fixation
 2. Where CO₂ combines with ribulose 1,5 – biphosphate
- (c) Explain the effect of high oxygen concentration on Rubisco and on photosynthesis. [2]
 1. Rubisco will bind oxygen instead of CO₂
 2. Glucose will not be produced through Calvin cycle
 3. Regeneration of RuBP will decrease
 4. Overall rate of photosynthesis will drop
- (d) Suggest why the leaves had to be frozen at the temperature of liquid nitrogen. [1]
To inactivate the enzymes, so that the enzyme activity can be measured accurately

In the same research, the cells from the different plants were studied using electron micrographs (shown in Fig. 4.2).

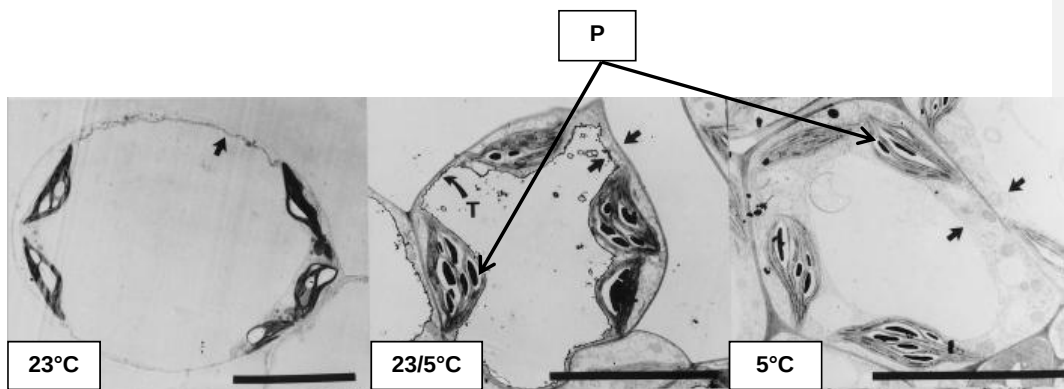


Fig. 4.2

- (e) Identify structure P. [1]

Starch grains

- (f) Describe the effect of temperature on the relative volume of cytoplasm and the relative volume of vacuole. [3]
1. A higher temperature of 23 °C results in a decrease/lowest in the relative volume of cytoplasm and an increase/highest relative volume of vacuole.
 2. A lower temperature of 5 °C results in an increase/highest in the relative volume of cytoplasm and a decrease/lowest relative volume of vacuole.
 3. A change in temperature from 23 to 5 °C results in a relative volume of cytoplasm and of vacuole being in between that of 23 °C and 5 °C.

[Total: 10]

END OF SECTION A

Section B: Essay Questions

- 5 (a) Describe how cellulose differs from starch. [6]

Structure:

1. Cellulose is made up of β glucose units while starch is made up of α glucose units
2. $\beta(1,4)$ glycosidic bond between glucose units in cellulose while $\alpha(1,4)$ glycosidic bond between glucose units in starch
3. In cellulose, alternate glucose units in a chain are rotated at 180° but glucose units are not rotated in the glucose chains in starch.
4. In cellulose, the chains remain straight, while in starch, each chain form a helix.
5. In cellulose, hydroxyl groups of each chain project outwards (thereby allowing cross-linking between chains) forming microfibril and macrofibril while in starch, hydroxyl groups projects inside the helix. (thereby no cross-linking between chains)

Function:

6. Cellulose has high tensile strength and functions as a structural material for plant cell wall. Starch being compact with a large number of glucose units functions well as a storage carbohydrate in plant cells.

- (b) Temperature is a factor that affects the permeability of living cells.
Using a beetroot that has red pigments in its cells, describe an investigation into the effect of temperature on the permeability of the cell. [9]

VARIABLES AND CONTROLLED VARIABLES

The independent variable is temperature.

The dependent variable is absorbance of solution.

DETAILED PROCEDURE

Note: All solutions are dispensed separately with clean and dry syringes and placed in clean and dry glassware (e.g. test tube and boiling tube) to prevent contamination or dilution.

1. Use a cork borer to obtain cylinders of fresh beetroot tissue.
2. Place a cylinder on a tile and hold it against a ruler with a forcep and cut them into 40 discs, each 3 mm thick with a scalpel.
3. Place all the discs in a small beaker and wash under a running tap. Start the stop watch.
4. After 5 minutes, stop the stopwatch and drained the water from the small beaker.
5. Take five of the beetroot discs and impale/pierce on a mounting needle with space between each disc.

6. Label seven test-tubes 20 °C, 30 °C, 40 °C, 50 °C, 60°C, 70 °C, 80 °C.
7. Add 10 ml distilled water to each tube with a 10ml syringe.
8. Set up a thermostatically-controlled water bath at 80 °C and place the test-tube labelled 80 °C in the water bath for 10 minutes. Start the stopwatch.
9. After 10 minutes, immerse the discs in the water bath for exactly 1 minute, then remove and carefully push the discs into the test-tube labelled 80 °C.
10. After the discs have stood for 30 minutes, shake gently to mix the contents of the tube and pour this liquid into a clean cuvette. Stop the stopwatch.
11. Fill a second cuvette with distilled water.
12. Place a blue filter into the colorimeter and use the distilled water to zero the machine adjusting the pointer to zero absorbance.
13. Measure and record the absorbance of the test solution (from step 10).
14. To ensure reliability of results, repeat steps 8 to 13 to obtain a total of 3 readings (triplicates) at this temperature using fresh samples of beetroot discs prepared from the same beetroot.
15. To show that temperature affects the amount of red pigment present in the solution, a control is set up. Steps 8 to 13 are performed with beetroot discs kept in water at room temperature (25°C). The control is subjected to the same environmental factors as that for the experiment.
16. Repeat steps 8 to 13 using the 6 other temperatures mentioned in step 3, i.e. 20°C, 30°C, 40°C, 50°C, 60°C and 70°C.
17. To ensure reproducibility of data, repeat the entire experiment (steps 1 to 14) twice using freshly prepared beetroot discs from a different beetroot plant.

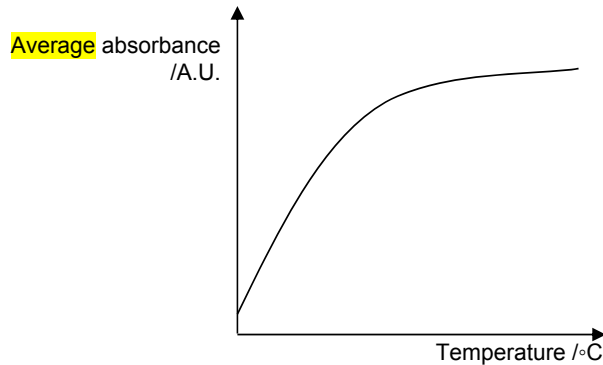
DATA MANIPULATION AND EXPECTED RESULTS

Table showing the absorbance value of the solution/ A.U. containing beetroot at different temperature /°C

Temperature /°C	Absorbance /arbitrary units			
	Reading 1	Reading 2	Reading 3	Average
20				
30				
40				
50				
60				
70				
80				

2. *[Draw a graph to show expected trends and/or results]*

Graph of average absorbance /arbitrary units against temperature /°C



3. *[Describe and Explain the expected trends and/or results]*

- As the temperature increases the average absorbance increases.
- This is because the concentration of red pigment in the beetroot increases.
- More red pigment leaves the cells as temperature increases due to an increase in membrane fluidity as the kinetic energy of the phospholipid molecules increases.
- Membrane permeability increases.
- At extreme temperature, the membrane structure would be disrupted and the membranes would be fully permeable.
- Maximum amount of red pigment can be found in the solution. Therefore average absorbance remains constant.

SAFETY PRECAUTIONS

1. Hold the plant material firmly on a flat surface with a forceps when cutting it to avoid being cut.
2. Handle cork borer and scalpel with care to avoid being cut.
3. Use a rag when handling the containers with hot water to prevent burns.

(c) Explain how uncontrolled cell division can result in cancer. [5]

1. When a cell undergoes division in the cell cycle, checkpoints in the cell cycle ensure that proper conditions must be achieved before a cell can proceed with the next stage of the cell cycle, and ultimately complete the cell division process.
2. Each checkpoint in the cell cycle is a critical control point where stop and go-ahead signals can regulate the cycle.
3. Normally, the regulatory proteins and factors ensure that cells grow and reproduce in an orderly and controlled way in each cell cycle.

4. And if the cell cycle goes wrong for any reason, the cell will not proceed beyond the checkpoints, and the cell ultimately dies.
 5. There are many regulatory proteins that control the cell cycle checkpoints such as Cyclin-dependent kinases (Cdks) and growth factors.
 6. Cdks function by switching on / off target proteins through phosphorylation, which will in turn be stimulated to phosphorylate other proteins, thereby transmitting the signals down the cell signal transduction pathways
 7. Growth factors are proteins that stimulate cell division. Upon binding to the cell surface receptors on the cell membrane, a series of molecular interactions leading to cell division, is triggered within the cell.
 8. However, in certain cells, the cell cycle goes wrong in a way that allows the cell to keep on dividing until a lump of cells called a tumour is formed. Such tumours can be benign or malignant
 9. Metastasis occurs when cancer cells from a tumour penetrates the blood or lymphatic system, travels through the bloodstream and invade normal tissues elsewhere in the body.
 10. Cancers can also spread through the body by the process of invasion. This occurs when cancer cells directly migrate to and penetrate into neighbouring tissues.
- 6 (a) Describe the properties of plasmids that allow them to be used for DNA cloning. [7]
1. A plasmid may contain a region called the multiple cloning site (MCS) which contains a number of unique restriction sites for the insertion of foreign DNA.
 2. The MCS is usually found within a genetic marker
 3. They all contain an origin of replication (ori) so that the vector can replicate itself and the inserted fragment inside the host cell.
 4. This allows INDEPENDENT replication of the vector and gene of interest,
 5. leading to the synthesis of multiple copies of the genes and the plasmids within a host cell.
 6. They may contain genetic markers (name an e.g. antibiotic resistance gene) which
 7. confer a well-defined phenotype on the host cell.
 8. These allow for selection of cells which have successfully taken up the vector.
Or
These allow for selection of transformed cells.
 9. They also allow one for identification of host cells which have taken up the recombinant DNA molecules.

10. Plasmids are small,
 11. hence they can carry the gene of interest into the bacterial cell more easily.

OR

10. Plasmids are circular,
 11. so they will not be degraded easily.
 12. Hence, genes inserted into plasmids are more likely to be expressed within the bacterial cell.
 13. Plasmids contain promoters that
 14. allow the inserted gene of interest to be transcribed once the plasmid is within the cell.

- (b) Discuss the social and ethical issues of the Human Genome Project.
 [8]

Social and Ethical Issues:

1a. (Clinical issue) Evaluation and regulation of genetic tests

- i.1. Genetic tests are used as tools to detect gene variants associated with a specific disease or condition, as well as for paternity testing and forensics.**

However there are issues regarding proper evaluation and regulation of genetic tests for accuracy and reliability of detection and diagnosis of diseases. Or

(b) There are also concerns if standards and quality-control measures are implemented in genetic testing procedures.

2.

Many genetic tests are marketed directly to consumers through the internet. This raises concerns because it leads to instances where customers receive clinically significant test results without appropriate counseling from a healthcare professional, and are left to interpret complex results and diagnoses on their own.

see 23andme.com)

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2b. (Clinical issue) Issue for Informed consent/decision

- (e)1. Proper counseling by certified medical professionals given to patients about the risks and limitations of genetic technology, whether it is genetic tests or gene therapy.**

OR

Proper counselling by certified medical professionals given to parents about the use of genetic information in reproductive decision making and reproductive rights.

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(ii)2. Therefore, there is a need for doctors and other healthcare professionals to be trained in the dissemination of genetic information and patient counselling in order to raise the standards of patient care

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(iii)3. This way, patients can make an informed decision about their own healthcare and how best to proceed.

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3c. Uncertainty issues associated with disease susceptibilities and complex conditions identified by genetic testing

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4.1. There is concern if genetic testing should be carried out for conditions linked to multiple genes / gene-environment interactions, e.g. heart disease, for which there is currently no treatment or cure.

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5.2. The knowledge of one's condition may have a negative impact on one's psychological / emotional well-being.

1.3. May lead to genetic discrimination

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(f) d. Reproductive issue

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9.1. There is also the concern if genetic testing of an unborn fetus for such conditions should be performed when the results might lead its parents to abort the pregnancy.

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e. 5. (Social issue) Genetic discrimination / stigmatization

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*1. With genetic information from HGP, there is a concern that there will be an emergence of a group of socially marginalized individuals that are deemed to be "genetically inferior".

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OR

3. Discrimination by insurance companies. Genetic information may be used by insurance companies to deny or limit insurance coverage for individuals who are genetically pre-disposed to future disease onset.

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Discrimination by employers:

* OR

1.2. Employers may choose to hire or retain only individuals who are not genetically pre-disposed to future disease onset as they are perceived to be more productive.

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OR

2. Employers can also turn away individuals who possess genes associated with certain behavioural traits (manic depression, obsessive-compulsive disorder, bipolar disorder etc.).

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3.

4.3.

(c) Describe the features of blood stem cells and explain their normal functions. [5]

Max marking

Features

1. Stem cells are capable of dividing and renewing themselves (self-renewal) for long periods.
 - a. It has the ability to go through numerous cycles of cell division while maintaining the undifferentiated state.
2. Stem cells are unspecialised.
 - a. A stem cell does not have a specialised function but it has the capacity to differentiate into a specialised cell that can carry out these functions.
3. Stem cells can give rise to specialised cells.
 - a. It gives rise to specialised cells through the process called differentiation.
4. Hematopoietic stem cells aka blood stem cells are found in bone marrow and in umbilical cord blood.
5. Blood stem cells are multipotent stem cells that have ability to differentiate into a limited range of cell type.
6. Present in both children and adult.

Functions:

- 1 The primary roles of blood stem cells in a living organism are to replace cells that have died or have lost their function in the tissue that they are found.
- 2 Give rise to all the blood cell types.
 - red blood cells
 - white blood cells (such as B lymphocytes, T lymphocytes, natural killer cells, neutrophils, basophils, eosinophils, monocytes, macrophages, etc)

[Total: 20]

END OF SECTION B